DEFINITION

A *service mesh* is an infrastructure layer that manages how microservices communicate with one another, handling tasks like routing, load balancing, service discovery, and security without requiring changes to the services themselves. Rather than bake these responsibilities into application code, the service mesh injects sidecar proxies alongside each service instance. These proxies intercept all inbound and outbound traffic, enabling the mesh to enforce policies, encrypt communication, and collect telemetry data transparently.

Each PEP deployment option shown in figures 4.3, 4.4, and 4.5 offers distinct trade-offs concerning flexibility, operational complexity, and enforcement consistency. Table 4.1 summarizes these differences to help you evaluate which pattern best fits a given system.